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3 October 1986
SUPERSEDING
MIL-C-0024643/36A(SH)
8 November 1985
MIL-C-24643/36
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3SWU (INCLUDING VARIATIONS LS3SWA AND LS3SWUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

This cable can be supplied in three variations: LS3SWU (unarmored), LS3SWA (armored) and LS3SWUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.023 inch. Colored insulation - one black, one white and one red conductor for each triad.
Fourth - Three conductors, one black, one white and one red, cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth - Binder tape over each triad, at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of AWG no. 34 or no. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, inner tape.
- Diameter over completed triad 0.240-inch nominal.

(B) *denotes changes.*

AMSC N/A FSC 6145
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Eighth - The required number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with triad no. 1, from the center outward. Fillers shall be employed to form a firm, round assembly.

Ninth - Binder tape applied helically with overlap.

Construction variant for type LS3SWU

Tenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS3SWA

Tenth - Cross-linked polyolefin jacket. (See table I for thickness).

Eleventh - Braided metal armor and paint.

Construction variant for type LS3SWUS

Tenth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

Eleventh - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

Thirteenth - Binder tape applied helically with overlap.

Fourteenth - Cross-linked polyolefin jacket. (See table I for special thickness). Cable surface marking required.

TABLE I. Details.

Military 1/ part no. M24643/36	Size 2/	Number of triads	Cable jacket thickness (inch) nominal	Overall diameter (inches)					
				Type LS3SWU		Type LS3SWA		Type LS3SWUS	
				min	max	min	max	min	max
-01UN	3	3	0.040	0.615	0.655	0.665	0.705	0.675	0.715
-02UN	7	7	.040	.880	.940	.930	.990	.940	1.000
-03UN	10	10	.050	1.100	1.180	1.150	1.230	1.160	1.240
-04UN	14	14	.050	1.200	1.280	1.250	1.330	1.260	1.340
-05UN	19	19	.050	1.370	1.450	1.420	1.500	1.430	1.510
-06UN	24	24	.050	1.640	1.760	1.690	1.790	1.700	1.800
-07UN	30	30	.050	1.760	1.860	1.810	1.910	1.820	1.920
-08UN	37	37	.060	1.890	1.990	1.910	2.010	1.950	2.050
-09UN	44	44	.060	2.140	2.240	2.190	2.290	2.200	2.300

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.

2/ Type LS3SWU, LS3SWA and LS3SWUS.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	6.80
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield (pair shields only).....	500
Insulation resistance - shield to shield (LS3SWUS only, overall shields only), volts, minimum.....	6.0
Megohms/1000 feet, minimum.....	1.0
Jacket flaws.....	No failure
Conductor and shield (LS3SWUS only) continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Hydrostatic (open end) - leakage at 300 lbf/in^2 in 2 hours, in^3, maximum.....	1.0
Capacitance	
Mutual capacitance - at 1 kHz, pF/ft, maximum.....	30
Capacitance unbalance - percent, maximum.....	5
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing (conductor) - cycles, minimum....	50

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EXAMINATION AND TESTS: (Continued)

	Requirements
Permanence of printing (jacket) (types LS3SWU and LS3SWUS only) cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Shield conformance to material, construction and coverage	No failure
Armor (type LS3SWA only) - conformance to material, construction and coverage	No failure
(B) Surface transfer impedance (type LS3SWUS only)	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-36)

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